

SHARK BAY AP, Australia

WMO: 954020

Lat: 25.892S

Lon: 113.577E

Elev: 34

StdP: 100.91

Time Zone: 8.00 (E08)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.7	9.8	-2.6	3.0	25.4	-0.5	3.6	24.3	9.7	19.5	8.9	19.1	3.5	120	0.556

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	11.0	39.1	21.0	37.2	21.0	35.2	21.0	24.8	32.1	24.1	31.2	23.4	30.4	6.3	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	17.8	27.8	22.1	16.8	27.2	21.3	16.0	26.7	75.8	32.5	72.9	31.1	70.1	30.5	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.0	11.2	10.1	DB	5.9	43.3	1.0	1.7	5.2	44.5	4.6	45.5	4.0	46.5	3.3	47.7	(4)
(5)			WB	4.5	25.6	1.0	0.8	3.8	26.2	3.2	26.6	2.7	27.1	2.0	27.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	22.4	27.3	28.0	27.2	24.3	20.8	17.7	16.4	17.5	19.0	21.3	23.7	25.5	(6)	
	DBStd	4.62	2.65	2.72	2.81	2.52	2.24	1.83	1.78	1.83	2.08	2.15	2.53	2.75	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	157	0	0	0	0	4	33	64	39	15	1	0	0	(9)	
	CDD10.0	4511	536	503	534	430	333	231	200	232	270	351	410	481	(10)	
	CDD18.3	1624	277	270	276	180	79	14	5	12	36	93	160	222	(11)	
	CDH23.3	14484	2703	2786	2749	1423	534	95	36	124	288	634	1213	1898	(12)	
	CDH26.7	6483	1323	1417	1371	577	138	7	2	18	75	201	489	866	(13)	
(14) Wind	WSAvg	5.8	7.2	6.9	6.3	5.5	4.6	4.3	4.2	4.6	5.4	6.4	6.9	7.4	(14)	
(15) Precipitation	PrecAvg	221	5	14	16	18	39	47	43	22	8	5	5	1	(15)	
	PrecMax	427	64	144	208	92	136	161	149	73	22	46	51	31	(16)	
	PrecMin	103	0	0	0	0	0	4	0	0	0	0	0	0	(17)	
	PrecStd	86	13	28	36	22	34	33	39	17	6	9	12	5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.6	41.8	40.5	36.7	32.5	27.6	26.4	29.0	31.6	34.9	37.9	40.1	(19)	
		MCWB	21.3	21.2	20.4	19.0	17.2	14.9	15.3	15.1	15.8	17.4	18.5	20.0	(20)	
	2%	DB	38.1	38.6	38.0	34.2	30.0	25.8	24.2	26.3	28.8	31.4	34.8	36.9	(21)	
		MCWB	21.8	21.7	20.9	18.8	16.7	15.1	14.9	14.8	15.7	17.3	18.9	20.4	(22)	
	5%	DB	35.4	36.0	35.8	32.1	28.0	24.3	22.7	24.5	26.5	28.6	31.8	33.9	(23)	
		MCWB	22.2	22.4	21.0	19.1	16.8	15.4	14.8	14.9	15.9	17.6	19.1	20.7	(24)	
	10%	DB	32.9	33.8	33.4	30.0	26.3	22.7	21.6	22.9	24.6	26.5	29.2	31.1	(25)	
		MCWB	22.2	22.8	21.1	19.3	16.8	15.2	14.6	14.9	15.7	17.5	19.4	20.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.4	26.1	24.7	23.5	22.2	20.5	19.6	19.2	19.9	20.6	22.2	23.9	(27)	
		MCDB	32.4	32.1	32.3	28.6	25.1	22.5	22.1	22.6	25.6	28.1	30.7	33.4	(28)	
	2%	WB	24.5	25.2	23.9	22.4	20.8	18.9	18.0	18.1	18.6	19.7	21.4	22.8	(29)	
		MCDB	31.7	32.2	30.9	27.4	24.2	21.5	21.0	21.6	24.1	26.4	28.9	31.2	(30)	
	5%	WB	23.8	24.5	23.2	21.6	19.8	17.8	16.9	17.3	17.7	19.0	20.7	22.1	(31)	
		MCDB	31.0	31.6	29.7	26.8	23.9	20.9	19.9	21.2	22.7	25.3	27.7	29.8	(32)	
	10%	WB	23.1	23.7	22.6	21.0	18.9	17.0	15.9	16.4	16.8	18.4	20.0	21.4	(33)	
		MCDB	30.4	30.6	28.9	26.2	23.0	20.3	19.4	20.7	21.9	24.2	26.7	28.2	(34)	
(35) Mean Daily Temperature Range	MDBR		10.9	11.0	11.2	11.2	11.5	10.7	10.9	11.7	11.7	11.3	10.9	10.9	(35)	
	5% DB	MCDBR	14.8	14.7	14.9	14.3	14.5	13.4	12.9	14.7	15.1	15.1	15.6	15.7	(36)	
		MCWBR	3.9	3.7	3.7	4.3	4.9	5.6	5.8	6.1	5.5	5.0	4.6	4.2	(37)	
	5% WB	MCDBR	11.2	11.1	11.0	11.0	10.3	9.1	10.2	11.8	12.0	11.5	12.1	11.8	(38)	
		MCWBR	3.4	3.5	3.3	3.8	4.9	5.4	6.3	6.5	5.4	4.4	4.2	3.7	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.363	0.369	0.351	0.330	0.310	0.304	0.297	0.301	0.318	0.328	0.336	0.343	(40)	
	taud		2.576	2.586	2.616	2.655	2.677	2.674	2.690	2.655	2.578	2.562	2.564	2.585	(41)	
	Ebn at Noon		979	959	947	925	904	889	910	942	966	988	1000	1001	(42)	
	Edh at Noon		107	104	96	84	75	72	73	83	97	105	108	106	(43)	
(44) All-Sky Solar Radiation	RadAvg		8.16	7.61	6.51	5.21	4.19	3.58	3.80	4.83	6.18	7.34	8.13	8.50	(44)	
	RadStd		0.34	0.24	0.41	0.27	0.24	0.18	0.23	0.18	0.18	0.21	0.19	0.29	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)		N/A	N/A	-1.75	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page